

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
book = pd.read_csv("data/BX-books.csv", sep=";", on_bad_lines='skip', encoding='latin-1')
```

C:\Users\Fakhur ali\AppData\Local\Temp\ipykernel_46372\2998565206.py:1: DtypeWarning: Columns (3) have mixed types. Specify dtype o
book = pd.read_csv("data/BX-books.csv", sep=";", on_bad_lines='skip', encoding='latin-1')

```
book.head()
```

	ISBN	Book-Title	Book-Author	Year-Of-Publication	Publisher	Image-URL-S
0	0195153448	Classical Mythology	Mark P. O. Morford	2002	Oxford University Press	http://images.amazon.com/images/P/0195153448.0...
1	0002005018	Clara Callan	Richard Bruce Wright	2001	HarperFlamingo Canada	http://images.amazon.com/images/P/0002005018.0...
2	0060973129	Decision in Normandy	Carlo D'Este	1991	HarperPerennial	http://images.amazon.com/images/P/0060973129.0...
3	0374157065	Flu: The Story of the Great Influenza Pandemic...	Gina Bari Kolata	1999	Farrar Straus Giroux	http://images.amazon.com/images/P/0374157065.0...
4	0393045218	The Mummies of Urumchi	E. J. W. Barber	1999	W. W. Norton & Company	http://images.amazon.com/images/P/0393045218.0...

```
book.shape
```

```
(271360, 8)
```

```
book.columns
```

```
Index(['ISBN', 'Book-Title', 'Book-Author', 'Year-Of-Publication', 'Publisher',  
      'Image-URL-S', 'Image-URL-M', 'Image-URL-L'],  
      dtype='object')
```

```
book.columns
```

```
Index(['ISBN', 'Book-Title', 'Book-Author', 'Year-Of-Publication', 'Publisher',  
      'Image-URL-S', 'Image-URL-M', 'Image-URL-L'],  
      dtype='object')
```

```
book = book[['ISBN', 'Book-Title', 'Book-Author', 'Year-Of-Publication', 'Publisher',  
            'Image-URL-L']]
```

```
book.head(2)
```

	ISBN	Book-Title	Book-Author	Year-Of-Publication	Publisher	Image-URL-L
0	0195153448	Classical Mythology	Mark P. O. Morford	2002	Oxford University Press	http://images.amazon.com/images/P/0195153448.0...

```
book.rename(columns={
    "Book-Title": "title",
    "Book-Author": "author",
    "Year-Of-Publication": "year",
    "Image-URL-L": "img_url"}, inplace=True)
```

```
book.columns
```

```
Index(['ISBN', 'title', 'author', 'year', 'Publisher', 'img_url'], dtype='object')
```

```
book.head(2)
```

	ISBN	title	author	year	Publisher	img_url
0	0195153448	Classical Mythology	Mark P. O. Morford	2002	Oxford University Press	http://images.amazon.com/images/P/0195153448.0...
1	0002005018	Clara Callan	Richard Bruce Wright	2001	HarperFlamingo Canada	http://images.amazon.com/images/P/0002005018.0...

now working on new file bx-user

```
users = pd.read_csv("data/BX-users.csv",sep=";", on_bad_lines='skip',encoding='latin-1')
```

```
users.head(2)
```

	User-ID	Location	Age
0	1	nyc, new york, usa	NaN
1	2	stockton, california, usa	18.0

```
users.shape
```

```
(278858, 3)
```

Ratings file

```
ratings = pd.read_csv("data/BX-book-ratings.csv",sep=";", on_bad_lines='skip',encoding='latin-1')
```

```
ratings.head(2)
```

	User-ID	ISBN	Book-Rating
0	276725	034545104X	0
1	276726	0155061224	5

```
ratings.shape
```

```
(1149780, 3)
```

```
print(book.shape)
print(users.shape)
print(ratings.shape)
```

```
(271360, 6)
(278858, 3)
(1149780, 3)
```

```
ratings.rename(columns={
    "User-ID" : "user_id",
    "Book-Rating" : "rating"},inplace=True)
```

```
ratings.head()
```

	user_id	ISBN	rating
0	276725	034545104X	0
1	276726	0155061224	5
2	276727	0446520802	0
3	276729	052165615X	3
4	276729	0521795028	6

```
ratings["user_id"].value_counts()
```

```

user_id
11676      13602
198711     7550
153662     6109
98391      5891
35859      5850
...
119573      1
276706      1
276697      1
276679      1
276676      1
Name: count, Length: 105283, dtype: int64

```

```
ratings["user_id"].unique().shape
```

```
(105283,)
```

```

x=ratings["user_id"].value_counts() >200
x

```

```

user_id
11676      True
198711     True
153662     True
98391      True
35859      True
...
119573    False
276706    False
276697    False
276679    False
276676    False
Name: count, Length: 105283, dtype: bool

```

```
x[x].shape
```

```
(899,)
```

```

y = x[x].index
y

```

```

Index([ 11676, 198711, 153662,  98391,  35859, 212898, 278418,  76352, 110973,
        235105,
        ...,
        116122,  44296,  28634,  59727,  73681, 274808, 188951,  9856, 155916,
        268622],
      dtype='int64', name='user_id', length=899)

```

```
ratings=ratings[ratings["user_id"].isin(y)]
```

```
ratings.head()
```

	user_id	ISBN	rating
1456	277427	002542730X	10
1457	277427	0026217457	0
1458	277427	003008685X	8
1459	277427	0030615321	0
1460	277427	0060002050	0

```
ratings.shape
```

```
(526356, 3)
```

```

ratings_with_book = ratings.merge(book, on = "ISBN")
ratings_with_book.head(3)

```

	user_id	ISBN	rating	title	author	year	Publisher	img_url
0	277427	002542730X	10	Politically Correct Bedtime Stories: Modern Ta...	James Finn Garner	1994	John Wiley & Sons Inc	http://images.amazon.com/images/P/002542730X.0...
1	277427	0026217457	0	Vegetarian Times Complete Cookbook	Lucy Moll	1995	John Wiley & Sons	http://images.amazon.com/images/P/0026217457.0...
2	277427	003008685X	8	Pioneers	James Fenimore Cooper	1974	Thomson Learning	http://images.amazon.com/images/P/003008685X.0...

```
num_rating = ratings_with_book.groupby('title')['rating'].count().reset_index()
```

```
num_rating.head(5)
```

	title	rating
0	A Light in the Storm: The Civil War Diary of ...	2
1	Always Have Popsicles	1
2	Apple Magic (The Collector's series)	1
3	Beyond IBM: Leadership Marketing and Finance ...	1
4	Clifford Visita El Hospital (Clifford El Gran...	1

```
num_rating.rename(columns={'rating' : 'num_of_rating'},inplace=True)
```

```
num_rating.head()
```

	title	num_of_rating
0	A Light in the Storm: The Civil War Diary of ...	2
1	Always Have Popsicles	1
2	Apple Magic (The Collector's series)	1
3	Beyond IBM: Leadership Marketing and Finance ...	1
4	Clifford Visita El Hospital (Clifford El Gran...	1

```
final_rating = ratings_with_book.merge(num_rating,on = 'title')
final_rating.head()
```

	user_id	ISBN	rating	title	author	year	Publisher	img_url	num_of_rating
0	277427	002542730X	10	Politically Correct Bedtime Stories: Modern Ta...	James Finn Garner	1994	John Wiley & Sons Inc	http://images.amazon.com/images/P/002542730X.0...	82
1	277427	0026217457	0	Vegetarian Times Complete Cookbook	Lucy Moll	1995	John Wiley & Sons	http://images.amazon.com/images/P/0026217457.0...	7
2	277427	003008685X	8	Pioneers	James Fenimore Cooper	1974	Thomson Learning	http://images.amazon.com/images/P/003008685X.0...	1
3	277427	003008685X	8	Ask for May, Settle for	G R	1974	Henry Holt	http://images.amazon.com/images/P/003008685X.0...	1

```
final_rating=final_rating[final_rating['num_of_rating']>50]
final_rating.head()
```

	user_id	ISBN	rating	title	author	year	Publisher	img_url	num_of_rati
0	277427	002542730X	10	Politically Correct Bedtime Stories: Modern Ta...	James Finn Garner	1994	John Wiley & Sons Inc	http://images.amazon.com/images/P/002542730X.0...	
13	277427	0060930535	0	The Poisonwood Bible: A Novel	Barbara Kingsolver	1999	Perennial	http://images.amazon.com/images/P/0060930535.0...	1
15	277427	0060934417	0	Bel Canto: A Novel	Ann Patchett	2002	Perennial	http://images.amazon.com/images/P/0060934417.0...	1
				One for the					

```
final_rating.sample(10)
```

	user_id	ISBN	rating	title	author	year	Publisher	img_url	num_of_ra
205478	115490	0451184963	0	Insomnia	Stephen King	1995	Signet Book	http://images.amazon.com/images/P/0451184963.0...	
320375	185384	0440211727	0	A Time to Kill	JOHN GRISHAM	1992	Dell	http://images.amazon.com/images/P/0440211727.0...	
419472	236283	0440236673	0	The Brethren	John Grisham	2000	Island	http://images.amazon.com/images/P/0440236673.0...	
199016	113817	0553582933	0	Winter Moon	Dean R. Koontz	2001	Bantam Books	http://images.amazon.com/images/P/0553582933.0...	
307932	178667	0451150694	0	Christine	Stephen King	1983	Signet Book	http://images.amazon.com/images/P/0451150694.0...	
154608	89602	0440224675	0	Hannibal	Thomas Harris	2000	Dell Publishing Company	http://images.amazon.com/images/P/0440224675.0...	
88787	49109	0316969680	7	The Beach House	James Patterson	2002	Little, Brown	http://images.amazon.com/images/P/0316969680.0...	
21074	44070	054540000X	7	The Reef	Nora	2000	Jove	http://images.amazon.com/images/P/054540000X.0...	

```
final_rating.drop_duplicates(['user_id', 'title'], inplace=True)
```

```
final_rating.shape
```

```
(57952, 9)
```

```
book_pivot = final_rating.pivot_table(columns='user_id', index='title', values='rating')
book_pivot
```

user_id	254	2276	2766	2977	3363	3757	4017	4385	6242	6251	...	274004	274061	274301	274308	274808	275970	277427
title																		
1984	9.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	0.0	NaN
1st to Die: A Novel	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2nd Chance	NaN	10.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	0.0	NaN	NaN	NaN
4 Blondes	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.0	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN
84 Charing Cross Road	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	10.0	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Year of Wonders	NaN	NaN	NaN	7.0	NaN	NaN	NaN	NaN	7.0	NaN	...	NaN	NaN	NaN	NaN	NaN	0.0	NaN
You Belong To Me	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Zen and the Art of Motorcycle Maintenance: An Inquiry into Values	NaN	NaN	NaN	NaN	0.0	NaN	NaN	NaN	NaN	0.0	...	NaN	NaN	NaN	NaN	NaN	0.0	NaN
Zoya	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN
\"Is for Outlaw"	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	8.0	NaN	NaN	NaN	NaN

703 rows × 888 columns

book_pivot.fillna(0,inplace=True)

book_pivot

user_id	254	2276	2766	2977	3363	3757	4017	4385	6242	6251	...	274004	274061	274301	274308	274808	275970	277427
title																		
1984	9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1st to Die: A Novel	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2nd Chance	0.0	10.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4 Blondes	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0
84 Charing Cross Road	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	10.0	0.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Year of Wonders	0.0	0.0	0.0	7.0	0.0	0.0	0.0	0.0	7.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0
You Belong To Me	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Zen and the Art of Motorcycle Maintenance: An Inquiry into Values	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Zoya	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0
\"Is for Outlaw"	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	8.0	0.0	0.0	0.0	0.0

703 rows × 888 columns

including new library scipy

```
from scipy.sparse import csr_matrix
```

```
book_sparse = csr_matrix(book_pivot)
```

```
book_sparse
```

```
<Compressed Sparse Row sparse matrix of dtype 'float64'
  with 14527 stored elements and shape (703, 888)>
```

```
from sklearn.neighbors import NearestNeighbors as nn
```

```
model = nn(algorithm='brute')
```

```
model.fit(book_sparse)
```

▼ NearestNeighbors ⓘ ?

► Parameters

```
distance , suggestion = model.kneighbors(book_pivot.iloc[237,:].values.reshape(1,-1),n_neighbors=6)
```

```
distance
```

```
array([[ 0.          , 22.53885534, 23.2594067 , 23.70653918, 23.72762104,
        23.9582971 ]])
```

```
suggestion
```

```
array([[237, 356, 296, 305, 377, 173]])
```

```
for i in range(len(suggestion)):
    print(book_pivot.index[suggestion[i]])
```

```
Index(['Hemlock Bay', 'No Safe Place', 'Last Man Standing',
       'Long After Midnight', 'Pleading Guilty', 'Exclusive'],
      dtype='object', name='title')
```

```
book_pivot.index[3]
```

```
'4 Blondes'
```

```
books_name = book_pivot.index
```

```
import pickle
pickle.dump(model , open('artifacts/model.pkl' , 'wb'))
pickle.dump(books_name , open('artifacts/books_name.pkl' , 'wb'))
pickle.dump(final_rating , open('artifacts/final_rating.pkl' , 'wb'))
pickle.dump(book_pivot , open('artifacts/book_pivot.pkl' , 'wb'))
```

```
def recommend_book(book_name):
    book_id = np.where(book_pivot.index == book_name)[0][0]
    distance , suggestion = model.kneighbors(book_pivot.iloc[book_id,:].values.reshape(1,-1),n_neighbors=6)

    for i in range(len(suggestion)):
        books = book_pivot.index[suggestion[i]]
        for j in books:
            print(j)
```

```
book_name= 'Hemlock Bay'
```

```
recommend_book(book_name)
```

```
Hemlock Bay
No Safe Place
Last Man Standing
Long After Midnight
```

Pleading Guilty
Exclusive

book_name= 'Pleading Guilty'

recommend_book(book_name)

Pleading Guilty
No Safe Place
Long After Midnight
Exclusive
Journey
Absolute Power

final_rating.head()

user_id	ISBN	rating	title	author	year	Publisher	img_url	num_of_rati
			Politically Correct	James		John Wiley		